THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

This exam paper must not be removed from the venue

Venue	____________________
Seat Number	__________
Student Number	|__|__|__|__|__|__|__|__|
Family Name	____________________
First Name	____________________

School of Electrical Engineering & Computer Science

Semester One Examinations, 2024

COMP2048 Theory of Computing

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

•No written or printed material permitted
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

None

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

Please use the space provided on this paper for your answers where possible. You should attempt all questions. You MUST show the steps used to arrive at your final solution. Best marks will be awarded to (numerically) correct solutions showing complete working. Partial credit will be given to partially complete solutions or to incorrect solutions where errors have been carried through calculations. Assumptions, if any, should be explicitly stated and justified/verified as appropriate. Best marks will be awarded to the ones most relevant to the course and the course material

For Examiner Use Only

Question	Mark

Total _________

Problem 1 (4 marks)

Provide a description of the incompleteness theorems ensuring that you:

a) Explain by use of examples what is a Gödel number and what is meant by Gödel numbering. Why is it useful and what insights does it provide in the context of mathematical systems?
b) Explain how the use of Gödel numbering and the method of proof by absurdum is used to prove the first incompleteness theorem.
c) Explain why feeding the Godel number of an expression into itself could cause a contradiction.
d) Explain how to prove the second incompleteness theorem with the use of the first incompleteness theorem.

Problem 2 (4 Marks)

All regular languages are closed under Union, Complement and Intersection. Define what it means for a language to be regular and utilise your knowledge of Finite State Machines (FSMs) to prove that regular languages have closure under Intersection.

Problem 3 (6 Marks)
Show that the language $C = \left\{\alpha^i \beta^j \gamma^k \;\middle|\; i \times j = k \text{ and } i, j, k \geq 1\right\}$ is not recognisable by a finite state machine by designing a Turing machine that decides it.

Problem 4 (4 Marks)
Recall that the XOR (Exclusive OR) operation is a logical operation that outputs true if and only if one of the inputs is true, but not both. Write a Lambda expression representing the XOR (Exclusive OR) operation. Define any helper lambda expressions used in your solution. Briefly explain your approach.

Problem 5 (6 Marks)

Describe how one can compute the factorial function using recursion in Lambda Calculus. Make sure to mention all the relevant combinators and theorems required to achieve your result.

Problem 6 (6 Marks)

Discuss the advantages and disadvantages of quantum computing as compared to classical computing. In your discussion provide:

a) An explanation of the quantum mechanics principles behind the operation of logic gates in quantum computers
b) Examples of problems uniquely addressed by quantum computers
c) A brief analysis of the theoretical reasons why quantum computers have the potential to outperform classical computers in problems
d) A brief analysis of the practical challenges that may undermine those theoretical advantages in practice.

Problem 7 (10 Marks)

Space agencies around the world have made an astonishing discovery of an alien civilisation many light years away in another part of our galaxy. The scientific research community is exploring which theories across science is most appropriate and universal (to allow ease of understanding) to communicate to the aliens. The Australian Space Agency (ASA) computer science team has decided to pitch to the community that computation is the most universal concept that should be communicated.

As part of the team, you have been tasked to put together a short proposal (approximately 1-page) that presents the most compact form of a theory of computing possible to demonstrate humanity's capability in this area to the alien civilisation. Contact with these aliens has been made via radio and radio telescopes, so the signal is severely limited. This means that the amount of information (effectively the number of characters/symbols) transmittable is very low and responses won't be received for hundreds of years. Thus, the signal to be sent must be a few short compact expressions that allow for arbitrary computation. This proposal must also include the actual signal that will be sent across vast distances via radio and will be subject to the limitations of the signals aforementioned.

Create this short proposal and design the signal to be sent to the alien civilisation ensuring that you:

a) Explain how arbitrary computations are made possible and its implications. (1 Mark)

b) Describe why computation is universal and may be considered as the most suitable theory to be sent to an alien civilisation over other scientific theories. (1 Mark)

c) Choose the appropriate theory of computation that is best suited for the task, i.e. the most compact to send over vast distances and will meet the above requirements of the signal. Justify your choice with an explanation of that theory and its relevant advantages that make it most suitable. (2 Marks)

d) Devise the necessary information and/or instruction set(s) necessary to communicate humanity's understanding of the theory of computation. Ensure to include any key equations, concepts and justifications for their inclusion. (3 Marks)

e) Write down the actual signal that will be sent, while providing your reasoning for its final form. Ensure that you provide the necessary information to represent computation and provide an example for the aliens to compute, so that they can confirm it is correct. (3 Marks)

Marks will be awarded based on the optimal solution to this problem. In other words, ensure your solution best addresses the problem.

END OF EXAMINATION